

CONTACT

- [GitHub](#)
- [LinkedIn](#)
- Reus, Spain

EDUCATION

- M.S. in International Business, Universitat Pompeu Fabra, 2025
- B.S. in Chemical Engineering, University of Southern California, 2008

PROFESSIONAL DEVELOPMENT

- **Microsoft Power BI / PL-300:** Power BI, DAX, data modeling and reporting
- **Google Data Analytics:** Data analysis, visualization, and analytical

methodology

- **Applied Machine Learning with Python:** Pandas, NumPy, Scikit-Learn and ML workflows
- **Full-Stack Web Development:** PHP, OOP, JavaScript, CSS, MySQL, REST APIs and Git
- **Google AI Professional Certificate:** Applied AI concepts and tools

SKILLS

- **BI & Visualization:** Power BI, DAX, Power Query, Advanced Excel, TIBCO Spotfire, Interactive Dashboards, KPI Development
- **Data Modeling & Architecture:** Kimball Dimensional Modeling, Star / Galaxy Schemas, Relational Schema Design, Data Transformation, Fact/Dimension Grain Alignment
- **Databases & Cloud:** SQL, MySQL, Microsoft Azure, ADLS Gen2, Azure Data Flows, Bicep IaC, Microsoft Entra ID, Private Endpoints
- **Programming & Full-Stack:** Python (Pandas, NumPy, Scikit-Learn), PHP (OOP / MVC), REST APIs, Git Version Control
- **Analytical & Business Methods:** Requirements Gathering, User Stories, Functional Specifications, Project Controls, Business Planning, KPI Development, Trend & Variance Analysis, Root-Cause Analysis, Performance Analysis, Cohort Analysis

Alejandra Badia

BUSINESS INTELLIGENCE DATA ANALYST | TECHNICAL PROJECT MANAGER

Data Analyst and BI Engineer backed by 11 years of engineering, systems measurement, and technical project management in the energy sector, a B.S. in Chemical Engineering, and a Master's in International Business. Proficient in Kimball dimensional modeling (star/galaxy schemas), advanced Power Query ETL, DAX measure engineering, relational SQL database design, and Azure cloud data flows. Background includes evaluating operational bottlenecks, auditing measurement and data integrity, analyzing system performance, and translating complex engineering workflows into decision-ready business intelligence.

EXPERIENCE

Web Development Practical Training | EXPACIA | Reus, Spain | 2025

- Designed structured JSON schemas and input-validation logic to convert user form data into standardized payloads for Unity ingestion
- Established a robust data contract between the frontend and game engine, ensuring data integrity and preventing downstream system errors

Advanced Education & Professional Upskilling | Career Break & Full-Time Professional Upskilling | 2020 – 2025

- Career break (2020–2023): Planned personal pause focused on family and international relocation
- Completed Master's in International Business and Full-Stack Web Development Professional Certificate (IFCD0210), with focus on PHP, JavaScript, SQL, and REST API integration
- Gained experience with Agile development workflows, Jira, Git collaboration, and iterative product prototyping

Technical Project Manager | CHEVRON | Lagos, Nigeria | 2012-2019

- Directed the full-lifecycle delivery of projects, including real-time monitoring and instrumentation systems, establishing reliable operational data flows and infrastructure across distributed mechanical and electrical domains
- Developed decision-support packages at key project phase gates, integrating project performance, technical analysis, cost, schedule, risk, and business considerations into recommendations for management review

- Presented project recommendations to Decision Review Boards and Decision Executives to support phase advancement, funding, scope, and execution decisions, securing required stakeholder sign-off
- Led cross-functional coordination with engineering, operations, construction and regulatory stakeholders to align system design with operational needs, managing project controls, schedules, cost and risk mitigations
- Supported annual business planning and project prioritization by evaluating operational requirements, resource needs, project performance, and delivery risks

Data & Measurement Systems Engineer | CHEVRON | Bakersfield, CA | 2010 – 2012

- Built analytical dashboards and data views using TIBCO Spotfire and Excel to evaluate manifold reliability, accuracy thresholds, and flow capacity for production engineering tie-in decisions
- Managed and audited automated well test (AWT) measurement systems across 300+ distributed production units, ensuring flow-rate measurement accuracy and instrument calibration
- Chaired weekly cross-functional operational review meetings with field teams and engineers to investigate measurement data anomalies, perform root-cause analysis, and prioritize corrective maintenance
- Built a predictive capacity and network model in Excel, integrating stakeholder production forecasts with structural P&ID data to run pipeline-by-pipeline bottleneck analysis across the distribution network
- Calculated granular, multi-scenario flow limits and financial cost estimates for each project package, presenting final data models to secure cross-functional engineering alignment and budget approval
- Spearheaded Lean/Six Sigma operational reliability initiatives, standardizing measurement error diagnostics and reducing Cost of Poor Quality (COPQ)

Process Engineer, Oil & Water Treatment Plants | CHEVRON | Bakersfield, CA | 2008-2010

- Defined end-to-end technical requirements and executed Reliability-Centered Maintenance (RCM) workflows to establish data-driven maintenance plans, improving asset uptime and operational reliability
- Led system validation and delivered technical analysis for Alarm Objective Analysis (AOA) and facility safety/compliance audits, reviewing system threshold triggers, enforcing Management of Change (MOC) protocols, and mitigating operational alarm fatigue
- Analyzed process and equipment performance data to identify operational trends, reliability issues, and improvement opportunities, translating findings into recommendations for maintenance optimization

SELECTED PROJECTS

E-Commerce Marketing BI Engine | Power BI, DAX, Power Query, Kimball Galaxy Schema

- Built an end-to-end BI solution integrating Shopify sales, GA4 traffic, and Google/Meta/TikTok advertising data across baseline and scaling periods
- Designed a Kimball galaxy schema and DAX measures to evaluate marketing efficiency, identifying a 253% increase in ad spend versus 23% Net Sales growth and an 89% decline in Marketing Profit

BI Measurement Planner (Kimball Spec Engine) | PHP, JavaScript, JSON, Markdown

- Developed a full-stack application that standardizes BI requirements, KPI definitions, data grain, dimensional modeling, and metric dependencies before development
- Automated cross-section data lineage and generated machine-readable specifications and data dictionaries for analyst/engineering handoff

Clinical Laboratory Operations Diagnostic Dashboard | Excel, Power Query, Power Pivot, DAX

- Built an interactive diagnostic dashboard analyzing 1,000 specimen runs to identify laboratory turnaround-time and data-integration bottlenecks
- Engineered relational model, SLA variance benchmarks, and dynamic DAX measures to isolate a 24.3-minute manual hold delay while confirming 0.0% IT/HL7 interface drops